

# Employability Skill Development

Unit I – Quantitative Aptitude: MCQs  
with Answers

# Numbers

## Types of Numbers

**Natural Numbers (N):** Counting numbers starting from 1.

Example: 1, 2, 3, 4, ...

**Whole Numbers (W):** Natural numbers including 0.

Example: 0, 1, 2, 3, ...

**Integers (Z):** Whole numbers and their negative counterparts.

Example: -3, -2, -1, 0, 1, 2, ...

**Rational Numbers (Q):** Numbers that can be expressed as  $p/q$ , where  $p$  and  $q$  are integers and  $q \neq 0$ .

Example:  $1/2$ ,  $-3/4$ , 5, 0.333...

**Irrational Numbers:** Cannot be expressed as  $p/q$ . Non-terminating, non-repeating decimals.

Example:  $\sqrt{2}$ ,  $\pi$

**Real Numbers (R):** Combination of rational and irrational numbers.

**Prime Numbers:** Numbers greater than 1 having exactly two factors: 1 and itself.

Example: 2, 3, 5, 7, 11...

**Composite Numbers:** Numbers having more than two factors.

Example: 4, 6, 8, 9...

**Even Numbers:** Multiples of 2.

**Odd Numbers:** Numbers not divisible by 2.

Properties:

$\text{Even} \pm \text{Even} = \text{Even}$

$\text{Odd} \pm \text{Odd} = \text{Even}$

$\text{Even} \times \text{Even} = \text{Even}$

$\text{Even} \times \text{Odd} = \text{Even}$

$\text{Odd} \times \text{Odd} = \text{Odd}$

**Perfect Numbers:** Equal to the sum of their proper divisors.

Example: 6 (1+2+3), 28 (1+2+4+7+14)

Which of the following is an even number?

a) 17

b) 42

c) 55

d) 91

Ans: B) 42

Sum of two odd numbers is always:

a) Odd

b) Even

c) Prime

d) Composite

Answer: b) Even

Which number is odd?

a) 126

b) 203

c) 118

d) 420

Answer: b) 203

Product of an even and an odd number is:

- a) Always odd
- b) Always even
- c) Sometimes odd
- d) Sometimes even

Answer: b) Always even

Which of these numbers is both odd and prime?

a) 27

b) 29

c) 35

d) 45

Answer: b) 29

If  $n$  is even, then  $n^2$  is:

- a) Even
- b) Odd
- c) Prime
- d) None

Answer: a) Even

Which of the following is an odd multiple of 7?

a) 28

b) 35

c) 42

d) 56

Answer: b) 35

Which set contains only even numbers?

a) {2, 4, 5, 8}

b) {6, 8, 12, 14}

c) {3, 6, 9, 12}

d) {1, 2, 3, 4}

Answer: b) {6, 8, 12, 14}

The sum of first 5 even numbers is:

a) 20

b) 25

c) 30

d) 35

Answer: c) 30

If an even number is subtracted from an odd number, the result is:

a) Even

b) Odd

c) Prime

d) Composite

Answer: b) Odd

# HCF (Highest Common Factor)

- HCF is the greatest number that divides two or more numbers exactly.
- Methods:
  1. Prime factorization – find common factors, multiply them.
  2. Division method – divide larger number by smaller until remainder = 0.
- Formula: Product of two numbers =  $\text{HCF} \times \text{LCM}$ .

- LCM is the smallest number that is exactly divisible by two or more numbers.

Methods to find LCM:

1. Prime factorization – take highest powers of all prime factors.
  2. Division method – divide numbers by common primes until all become 1.
- Formula: Product of two numbers =  $\text{LCM} \times \text{HCF}$

# Numbers – Q1

HCF of 8 and 12 is:

A) 2

B) 4

C) 6

D) 12

Ans: B) 4

HCF of 15 and 25 is:

A) 3

B) 4

C) 5

D) 10

Ans: C) 5

HCF of 16, 24, 36 is:

A) 2

B) 4

C) 8

D) 12

Ans: B) 4

If HCF of two numbers is 13 and their product is 2028, their LCM is:

A) 132

B) 156

C) 172

D) 108

Ans: B) 4

HCF of  $2^3 \times 3^2 \times 5$  and  $2^2 \times 3^3 \times 7$  is:

A)  $2^2 \times 3^2$

B)  $2^3 \times 3^3$

C)  $2^2 \times 3^3$

D)  $2^3 \times 3^2$

Ans: A)  $2^2 \times 3^2 = 36$

LCM of 4 and 6 is:

A) 8

B) 12

C) 16

D) 24

Ans: B) 12

LCM of 5, 10, and 15 is:

A) 15

B) 30

C) 60

D) 90

Ans: B) 30

If  $\text{HCF} = 4$ ,  $\text{LCM} = 60$ , find product of numbers.

A) 240

B) 120

C) 180

D) 2400

Ans: A) 240

LCM of  $2^2 \times 3^3 \times 5$  and  $2^3 \times 3^2 \times 7$  is:

A)  $2^3 \times 3^3 \times 5 \times 7$

B)  $2^2 \times 3^3 \times 7$

C)  $2^3 \times 3^2 \times 5 \times 7$

D)  $2^3 \times 3^3 \times 7$

Ans: A)  $2^3 \times 3^3 \times 5 \times 7 = 2520$

If two numbers are coprime, their LCM is always:

A) Sum

B) Difference

C) Product

D) HCF

Ans: C) Product

Percentage means “per hundred”  $\rightarrow x\% = x/100$ .

25% of 200 is:

A) 25

B) 40

**C) 50**

D) 100

Q2. 10% of a number is 25. The number is:

A) 150

B) 200

**C) 250**

D) 300

A price increases from ₹200 to ₹250.  
Percentage increase is:

A) 20%

**B) 25%**

C) 30%

D) 35%

Q4. If 40% of  $x = 200$ , then  $x = ?$

A) 400

**B) 500**

C) 600

D) 800

A student scores 45 marks out of 60.  
Percentage = ?

A) 65%

B) 70%

**C) 75%**

D) 80%

A shopkeeper makes a profit of 15% on an item sold for ₹460. Find CP.

**A) ₹400**

B) ₹390

C) ₹360

D) ₹350

A quantity is increased by 20% and then decreased by 20%. Net % change = ?

A) 0%

B) 2% decrease

**C) 4% decrease**

D) 4% increase

The population of a town increases by 10% annually. If it is 10,000 now, what will it be after 2 years?

A) 11,000

B) 12,000

**C) 12,100**

D) 12,210

A's salary is 25% more than B's. By what % is B's less than A's?

**A) 20%**

B) 25%

C) 30%

D) 33.33%

The difference of two numbers is 1365. On dividing the larger number by the smaller, we get 6 as quotient and the 15 as remainder. What is the smaller number ?

A) 260

**B) 270**

C) 280

D) 300

If the number  $517*324$  is completely divisible by 3, then the smallest whole number in the place of \* will be:

A) 0

B) 1

**C) 2**

D) 3

The smallest 3 digit prime number is:

**A) 101**

B) 103

C) 109

D) 113

Which one of the following numbers is exactly divisible by 11?

A) 235641

B) 245642

C) 315624

**D) 415624**

The sum of first 45 natural numbers is:

A) 990

B) 1005

**C) 1035**

D) 1045

The difference between the local value and the face value of 7 in the numeral 32675149 is

A) 69,973

**B) 69,993**

C) 69,997

D) 69,999

A 3-digit number  $4a3$  is added to another 3-digit number  $984$  to give a 4-digit number  $13b7$ , which is divisible by  $11$ . Then,  $(a + b) = ?$

A) 4

B) 5

**C) 10**

D) 7

In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 46, what is the dividend ?

A) 5236

**B) 5336**

C) 5436

D) 5536

In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs?

**A) 6.25**

B) 6.5

C) 6.75

D) 7

A family consists of two grandparents, two parents and three grandchildren. The average age of the grandparents is 67 years, that of the parents is 35 years and that of the grandchildren is 6 years. What is the average age of the family.

**A) 31.71 years**

B) 32.50 years

C) 30.00 years

D) 34.50 years

The average monthly income of P and Q is Rs. 5050. The average monthly income of Q and S is Rs. 6250 and the average monthly income of P and S is Rs. 5200. The monthly income of P is:

A) Rs. 5000

**B) Rs. 4000**

C) Rs. 4500

D) Rs. 4200

The average age of husband, wife and their child 3 years ago was 27 years and that of wife and the child 5 years ago was 20 years. The present age of the husband is:

A) 35 years

**B) 40 years**

C) 45 years

D) 42 years

If the average marks of three batches of 55, 60 and 45 students respectively is 50, 55, 60, then the average marks of all the students is:

A) 54.50

**B) 54.69**

C) 55.00

D) 53.75

Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is:

A) 3 : 4

**B) 4 : 5**

C) 5 : 6

D) 2 : 3

A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1000 more than D, what is B's share?

A) Rs. 1000

B) Rs. 1500

**C) Rs. 2000**

D) Rs. 2500

Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?

A) 7 : 10.5 : 14

**B) 14 : 21 : 28**

C) 10 : 15 : 20

D) 21 : 28 : 35

Salaries of Ravi and Sumit are in the ratio 2 : 3.  
If the salary of each is increased by Rs. 4000,  
the new ratio becomes 40 : 57. What is Sumit's  
salary?

A) Rs. 32000

**B) Rs. 34000**

C) Rs. 36000

D) Rs. 30000

The salaries A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?

**A) 23 : 33 : 60**

B) 22 : 33 : 55

C) 20 : 30 : 50

D) 25 : 35 : 60

The fourth proportional to 5, 8, 15 is:

A) 20

**B) 24**

C) 25

D) 30

Two number are in the ratio 3 : 5. If 9 is subtracted from each, the new numbers are in the ratio 12 : 23. The smaller number is:

A) 27

B) 30

**C) 33**

D) 35

In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there?

A) 100

B) 120

**C) 150**

D) 180

A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:

A) Rs. 1425

**B) Rs. 1500**

C) Rs. 1537.50

D) Rs. 1576

Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?

A) 5 : 7 : 8

**B) 20 : 49 : 64**

C) 38 : 28 : 21

D) None of these

Simran started a software business by investing Rs. 50,000. After six months, Nanda joined her with a capital of Rs. 80,000. After 3 years, they earned a profit of Rs. 24,500. What was Simran's share in the profit?

- A) Rs. 9,423
- B) Rs. 10,250
- C) Rs. 12,500
- D) Rs. 10,500**

A and B started a business in partnership investing Rs. 20,000 and Rs. 15,000 respectively. After six months, C joined them with Rs. 20,000. What will be B's share in total profit of Rs. 25,000 earned at the end of 2 years from the starting of the business?

**A) Rs. 7500**

B) Rs. 9000

C) Rs. 9500

D) Rs. 10,000

A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is :

**A)  $\frac{8}{15}$**

B)  $\frac{7}{15}$

C)  $\frac{11}{15}$

D)  $\frac{2}{5}$

A can do a piece of work in 4 hours; B and C together can do it in 3 hours, while A and C together can do it in 2 hours. How long will B alone take to do it?

A) 8 hours

B) 10 hours

**C) 12 hours**

D) 24 hours

A and B can together finish a work 30 days. They worked together for 20 days and then B left. After another 20 days, A finished the remaining work. In how many days A alone can finish the work?

- A) 40 days
- B) 50 days
- C) 54 days
- D) 60 days**

Three pipes A, B and C can fill a tank from empty to full in 30 minutes, 20 minutes, and 10 minutes respectively. When the tank is empty, all the three pipes are opened. A, B and C discharge chemical solutions P, Q and S respectively. What is the proportion of the solution S in the liquid in the tank after 3 minutes?

A)  $\frac{5}{11}$

**B)  $\frac{6}{11}$**

C)  $\frac{7}{11}$

D)  $\frac{8}{11}$

Two pipes can fill a tank in 20 and 24 minutes respectively and a waste pipe can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is:

- A) 60 gallons
- B) 100 gallons
- C) 120 gallons**
- D) 180 gallons

A tank is filled by three pipes with uniform flow. The first two pipes operating simultaneously fill the tank at the same time during which the tank is filled by the third pipe alone. The second pipe fills the tank 5 hours faster than the first pipe and 4 hours slower than the third pipe. The time required by the first pipe is:

- A) 6 hours
- B) 10 hours
- C) 15 hours**
- D) 30 hours

A car covers 120 km at a speed of 40 km/h and the next 120 km at a speed of 60 km/h. The average speed of the car is:

**A) 48 km/h**

B) 50 km/h

C) 52 km/h

D) 54 km/h

A man walks at 6 km/h and runs at 12 km/h. If the total distance is 72 km and he spends half the time walking and half the time running, how long does he take?

A) 6 hours

**B) 8 hours**

C) 10 hours

D) 12 hours

A train 180 m long passes a man standing on a platform in 12 seconds. What is the speed of the train?

A) 45 km/h

B) 50 km/h

**C) 54 km/h**

D) 60 km/h

Two trains of lengths 120 m and 180 m are moving in opposite directions at speeds of 42 km/h and 48 km/h. The time taken to cross each other is:

**A) 12 sec**

B) 18 sec

C) 20 sec

D) 24 sec

A cyclist goes from town A to town B at 12 km/h and returns at 8 km/h. What is his average speed for the whole journey?

A) 9.2 km/h

B) 9.5 km/h

**C) 9.6 km/h**

D) 10 km/h

A train running at the speed of 60 km/hr crosses a pole in 9 seconds. What is the length of the train?

A)120m

B)180m

C)324m

**D)150m**

Two trains are moving in opposite directions @ 60 km/hr and 90 km/hr. Their lengths are 1.10 km and 0.9 km respectively. The time taken by the slower train to cross the faster train in seconds is:

A)36

B)45

**C)48**

D)49

The length of the bridge, which a train 130 metres long and travelling at 45 km/hr can cross in 30 seconds, is:

A) 200 m

**B) 225 m**

C) 245 m

D) 250 m

A boat can travel with a speed of 13 km/hr in still water. If the speed of the stream is 4 km/hr, find the time taken by the boat to go 68 km downstream.

A) 2 hours

B) 3 hours

**C) 4 hours**

D) 5 hours

A boat running downstream covers a distance of 16 km in 2 hours while for covering the same distance upstream, it takes 4 hours. What is the speed of the boat in still water?

A) 4 km/hr

**B) 6 km/hr**

C) 8 km/hr

D) Data inadequate

The speed of a boat in still water is 15 km/hr and the rate of current is 3 km/hr. The distance travelled downstream in 12 minutes is:

1.2 km

1.8 km

2.4 km

**3.6 km**

Tea worth Rs. 126 per kg and Rs. 135 per kg are mixed with a third variety in the ratio 1 : 1 : 2. If the mixture is worth Rs. 153 per kg, the price of the third variety per kg will be:

A) Rs. 169.50

B) Rs. 170

**C) Rs. 175.50**

D) Rs. 180

A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the containers so as to get 12 litres of milk such that the ratio of water to milk is 3 : 5?

A) 4 litres, 8 litres

**B) 6 litres, 6 litres**

C) 5 litres, 7 litres

D) 7 litres, 5 litres

A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is:

A)  $1/3$

**B)  $2/3$**

C)  $2/5$

D)  $3/5$

A jar contains 30% salt solution. A part of this is replaced with another solution containing 12% salt. The final mixture has 21% salt. What fraction of the solution was replaced?

- A)  $1/3$
- B)  $2/3$
- C)  $3/5$
- D)  $1/2$**

The cost of Type 1 rice is Rs. 15 per kg and Type 2 rice is Rs. 20 per kg. If both Type 1 and Type 2 are mixed in the ratio of 2 : 3, then the price per kg of the mixed variety of rice is:

**A) Rs. 18**

B) Rs. 18.50

C) Rs. 19

D) Rs. 19.50

How many kilogram of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per kg so that there may be a gain of 10% by selling the mixture at Rs. 9.24 per kg?

- 36 kg
- 42 kg
- 54 kg
- **63 kg**

Find the ratio in which rice at Rs. 7.20 a kg be mixed with rice at Rs. 5.70 a kg to produce a mixture worth Rs. 6.30 a kg.

A) 1 : 3

B) 2 : 3

**C) 3 : 2**

D) 4 : 5